Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – July 11th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. DOALL loops have linear speedup even using the Amdahl law. Could you please write a short pseudo code showing an example of a DOALL loop? (2)

B. Is the following code parallelizable? Are there better parallelizable solutions? (2)

```
for (i=0; i<100; i++)
        for (j=0; j<91; j++)
            a[i][j+9] = f(a[i][j]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the stencil pattern: assumptions, complexity, and example. (2)

B. Please describe how the Pack parallel pattern could be implemented. (1)

C. What is meant by "*permutation scatter*"? (1)

Exercise n. 3

Answer the following questions about OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe what is meant by pipelined parallelism and how could be implemented in OpenMP?(2)

B. Please describe the work-sharing constructs in OpenMP. (2)

Exercise n. 4

Answer the following questions about MPI and domain-specific languages and briefly explain (without explanation, the answer will be considered invalid)

A. Describe the functionality and the parameters meaning of the function MPI_Bcast (You can skip this question if you completed the second challenge) (2)

B. Please shortly describe the main characteristic of the domain-specific language Halide. (2)